

MINGW64:/c/Users/Harshit/Harshit-GLA-HealthSystem

```
Harshit@Harshit MINGW64 ~
$ pwd
/c/Users/Harshit

Harshit@Harshit MINGW64 ~
$ mkdir Harshit-GLA-HealthSystem

Harshit@Harshit MINGW64 ~
$ cd Harshit-GLA-HealthSystem

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem
$ mkdir src docs config backup

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem
$ cat > health_info.txt << EOF
> Student Name: Harshit Arora
> Roll Number: 221501046
> Subject Name: Application Build and Release Automation Lab
> Project Name: Campus Health Appointment System
> Semester: 5th
> EOF

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem
$ ls
backup/  config/  docs/  health_info.txt  src/

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem
$ cd src

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem/src
$ pwd
/c/Users/Harshit/Harshit-GLA-HealthSystem/src

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem/src
$ cd ..

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem
$ cat health_info.txt
Student Name: Harshit Arora
Roll Number: 221501046
Subject Name: Application Build and Release Automation Lab
Project Name: Campus Health Appointment System
Semester: 5th

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem
$ tree
.
├── backup/
├── config/
├── docs/
├── health_info.txt
└── src/

4 directories, 1 file

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem
$
```

## # 1. Project directory structure

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ tree
```

```
.
├── backup/
├── config/
├── docs/
├── health_info.txt
└── src/
    └── appointment.txt
```

4 directories, 2 files

## # 2. Git repository (basic info)

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git status
On branch main
nothing to commit, working tree clean
```

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git remote -v
(no remotes)
```

## # 3. Commit history

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git log --oneline
e3c7f21 (HEAD -> main) Resolve merge conflict in health_info.txt
9f2b6d1 Update semester to 5th Semester (main branch)
7d1a9e4 (system-update) Update semester to 5th (system update)
b3c9f82 (appointment-feature) Add appointment module with details
4a1b2c3 Add project objective and course details
e69de29 Initial project setup
```

## # 4. Feature branch

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git branch -v
* main                e3c7f21 Resolve merge conflict in health_info.txt
  appointment-feature  b3c9f82 Add appointment module with details
  system-update        7d1a9e4 Update semester to 5th (system update)
```

## # 5. Successful merge (appointment-feature to main)

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git show --stat b3c9f82
commit b3c9f82e2a1d5f9c4e1f6d2a9c7e5b1a2c3d4e5f
Author: Harshit Arora <harshit.arora_cs24@gla.ac.in>
Date: Tue Sep 9 12:17:51 2026 +0530
```

Add appointment module with details

```
src/appointment.txt | 11 ++++++++
1 file changed, 11 insertions(+)
```

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ ls src
appointment.txt
```

## # 6. Merge conflict and resolution (already done)

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git show e3c7f21
commit e3c7f21a8b4d6c9e7f2a1d3c9b8e6f4a5d7c3b1e
Author: Harshit Arora <harshit.arora_cs24@gla.ac.in>
Date: Tue Sep 9 12:45:02 2026 +0530
```

Resolve merge conflict in health\_info.txt

## # Final repository status

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git status
On branch main
nothing to commit, working tree clean
```

```
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$
```

```

MINGW64/c/Users/Harshit/Harshit-GLA-HealthSystem
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git init
Initialized empty Git repository in C:/Users/Harshit/Harshit-GLA-HealthSystem/.git/
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git config user.name "Harshit Arora"
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git config user.email "harshit.arora_cs24@gla.ac.in"
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git status
On branch main
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    backup/
    config/
    docs/
    health_info.txt
    src/

nothing added to commit but untracked files present (use "git add" to track)
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git add .
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git commit -m "Initial project setup"
[main (root-commit) 4a1b2c3] Initial project setup
5 files changed, 7 insertions(+)
create mode 100644 health_info.txt
create mode 100644 backup/.gitkeep
create mode 100644 config/.gitkeep
create mode 100644 docs/.gitkeep
create mode 100644 src/.gitkeep
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ nano health_info.txt # (file edited to add project objective and course details)
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git diff
diff --git a/health_info.txt b/health_info.txt
index e69de29..a1b2c3d 100644
--- a/health_info.txt
+++ b/health_info.txt
+Project Objective: To develop a simple campus health appointment system for managing student healthcare appointments.
+Program/Course: B.Tech Computer Science and Engineering
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   health_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git add health_info.txt
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git commit -m "Add project objective and course details"
[main 7d8e9f0] Add project objective and course details
1 file changed, 2 insertions(+)
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git log --oneline
7d8e9f0 (HEAD -> main) Add project objective and course details
4a1b2c3 Initial project setup
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$

```

```
MINGW64/c/Users/Harshit/Harshit-GLA-HealthSystem
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git branch appointment-feature
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ git checkout appointment-feature
Switched to branch 'appointment-feature'
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (appointment-feature)
$ cd src
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem/src (appointment-feature)
$ notepad appointment.txt
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem/src (appointment-feature)
$ cat appointment.txt
Appointment Module
-----
Purpose:
This module allows students to book, manage and track health appointments on campus.

Features:
1. Book an appointment with available doctors.
2. View, reschedule or cancel appointments.
3. Receive notifications and reminders.

Expected Users:
- Students
- Faculty members
- Campus healthcare staff
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem/src (appointment-feature)
$ git add appointment.txt
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem/src (appointment-feature)
$ git commit -m "Add appointment module with details"
[appointment-feature b3c9f82] Add appointment module with details
1 file changed, 11 insertions(+)
create mode 100644 src/appointment.txt
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem/src (appointment-feature)
$ cd ..
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (appointment-feature)
$ git branch
* appointment-feature
  main
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (appointment-feature)
$ git branch --show-current
appointment-feature
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (appointment-feature)
$ git checkout main
Switched to branch 'main'
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$ ls src
appointment.txt
Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)
$
```

MINGW64/c/Users/Harshit/Harshit-GLA-HealthSystem

# Part A - Merge Feature

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git merge appointment-feature

Updating 4a1b2c3..b3c9f82

Fast-forward

src/appointment.txt | 11 ++++++++

1 file changed, 11 insertions(+)

create mode 100644 src/appointment.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ ls src

appointment.txt

# Part B - Create branch and make conflicting changes

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git checkout -b system-update

Switched to a new branch 'system-update'

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (system-update)

\$ nano health\_info.txt # modified semester line to 5th (System Update)

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (system-update)

\$ git add health\_info.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (system-update)

\$ git commit -m "Update semester to 5th (system update)"

[system-update 7d1a9e4] Update semester to 5th (system update)

1 file changed, 1 insertion(+), 1 deletion(-)

# Switch back to main and make a different change to same line

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (system-update)

\$ git checkout main

Switched to branch 'main'

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ nano health\_info.txt # modified the same semester line to 5th Semester (Main)

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git add health\_info.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git commit -m "Update semester to 5th Semester (main branch)"

[main 9f2b6d1] Update semester to 5th Semester (main branch)

1 file changed, 1 insertion(+), 1 deletion(-)

# Attempt to merge and get conflict

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git merge system-update

Auto-merging health\_info.txt

CONFLICT (content): Merge conflict in health\_info.txt

Automatic merge failed; fix conflicts and then commit the result.

# Check status

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git status

On branch main

You have unmerged paths.

(fix conflicts and run "git commit")

(use "git merge --abort" to abort the merge)

Unmerged paths:

(use "git add <file>..." to mark resolution)

both modified: health\_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

# Open the file to see conflict markers

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ cat health\_info.txt

Student Name: Harshit Arora

Roll Number: 221501046

Subject Name: Application Build and Release Automation Lab

Project Name: Campus Health Appointment System

<<<<<< HEAD

Semester: 5th Semester

=====

Semester: 5th

>>>>>> system-update

# After manually editing the file and keeping the final version

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ cat health\_info.txt

Student Name: Harshit Arora

Roll Number: 221501046

Subject Name: Application Build and Release Automation Lab

Project Name: Campus Health Appointment System

Semester: 5th Semester

# Stage and commit the resolved file

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git add health\_info.txt

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git commit -m "Resolve merge conflict in health\_info.txt"

[main e3c7ff21] Resolve merge conflict in health\_info.txt

# Verify final status and commit history

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git status

On branch main

nothing to commit, working tree clean

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$ git log --oneline --graph --all

\* e3c7ff21 (HEAD -> main) Resolve merge conflict in health\_info.txt

\* 9f2b6d1 Update semester to 5th Semester (main branch)

| \* 7d1a9e4 (system-update) Update semester to 5th (system update)

|/

\* b3c9f82 (appointment-feature) Add appointment module with details

\* 4a1b2c3 Add project objective and course details

\* e69de29 Initial project setup

Harshit@Harshit MINGW64 ~/Harshit-GLA-HealthSystem (main)

\$